

Last name _____

First name _____

LARSON—MATH 756—CLASSROOM WORKSHEET 09
Modeling Combinatorial Problems with Integer Programs

Review

1. Why is the “hard” direction of Hall’s Theorem true? (and what is the main proof idea)?
2. What is König’s Theorem?
3. What is a *Min-Max Theorem*? What is an advantage of a min-max theorem?
4. What is the “easy” direction of König’s Theorem?

New

1. Why is the “hard” direction of König’s Theorem true? (and what is the main proof idea)?
2. What is a Vertex Packing Integer Program?
3. What are examples?

4. What is a Vertex Packing Linear Program? (and what is α_f ?)
5. What are examples?
6. What is important or significant about linear programs?
7. Why is $\alpha \leq \alpha_f$ for any graph?
8. What other constraints could we add to our IP (and LP) models?
9. What is Balinski's Lemma?